

Foundations of Semantic segmentation

[Semantic segmentation]

1.0 Content Block

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

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Graph partitioning methods Graph partitioning methods are an effective tools for image segmentation since they model the impact of pixel neighborhoods on a given cluster of pixels or pixel, under the assumption of homogeneity in images. In these methods, the image is modeled as a weighted, undirected graph. Usually a pixel or a group of pixels are associated with nodes and edge weights define the (dis)similarity between the neighborhood pixels. The graph (image) is then partitioned according to a criterion designed to model "good" clusters. Each partition of the nodes (pixels) output from these algorithms are considered an object segment in the image; see Segmentation-based object categorization. Some popular algorithms of this category are normalized cuts, random walker, minimum cut, isoperimetric partitioning, minimum spanning tree-based segmentation, and segmentation-based object categorization.

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Application of Isolated Point Detection in X-ray Image Processing The detection of isolated points has significant applications in various fields, including X-ray image processing. For instance, an original X-ray image of a turbine blade can be examined pixel-by-pixel to detect porosity in the upper-right quadrant of the blade. The result of applying an edge detector's response to this X-ray image can be approximated. This demonstrates the segmentation of isolated points in an image with the aid of single-pixel probes.

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This above equation is used to determine whether a point in the image is an isolated point based on the response magnitude $|R(x, y)|$ and a threshold value T . If the response magnitude is greater than or equal to the threshold, the function returns 1, indicating the presence of an isolated point; otherwise, it returns 0. This helps in the effective detection and segmentation of isolated points in the image.

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Histogram-based methods Histogram-based methods are very efficient compared to other image segmentation methods because they typically require only one pass through the pixels. In this technique, a histogram is computed from all of the pixels in the image, and the peaks and valleys in the histogram are used to locate the clusters in the image. Color or intensity can be used as the measure. A refinement of this technique is to recursively apply the histogram-seeking method to clusters in the image in order to divide them into smaller clusters. This operation is repeated with smaller and smaller clusters until no more clusters are formed. One disadvantage of the histogram-seeking method is that it may be difficult to identify significant peaks and valleys in the image. Histogram-based approaches can also be quickly

adapted to apply to multiple frames, while maintaining their single pass efficiency. The histogram can be done in multiple fashions when multiple frames are considered. The same approach that is taken with one frame can be applied to multiple, and after the results are merged, peaks and valleys that were previously difficult to identify are more likely to be distinguishable. The histogram can also be applied on a per-pixel basis where the resulting information is used to determine the most frequent color for the pixel location. This approach segments based on active objects and a static environment, resulting in a different type of segmentation useful in video tracking.

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$$\frac{\partial^2 f(x, y)}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y) \quad \{\displaystyle \frac{\partial^2 f(x,y)}{\partial x^2}\}=f(x+1,y)+f(x-1,y)-2f(x,y)\}$$

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the algorithm selects the newly formed graph. Simulated annealing requires the input of temperature schedules which directly affects the speed of convergence of the system, as well as energy threshold for minimization to occur.

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Image segmentation using MAP and expectation–maximization The expectation–maximization algorithm is utilized to iteratively estimate the a posterior probabilities and distributions of labeling when no training data is available and no estimate of segmentation model can be formed. A general approach is to use histograms to represent the features of an image and proceed as outlined briefly in this three-step algorithm: 1. A random estimate of the model parameters is utilized. 2. E step: Estimate class statistics based on the random segmentation model defined. Using these, compute the conditional probability of belonging to a label given the feature set is calculated using naive Bayes' theorem.

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Variational methods The goal of variational methods is to find a segmentation which is optimal with respect to a specific energy functional. The functionals consist of a data fitting term and a regularizing terms. A classical representative is the Potts model defined for an image f $\{\displaystyle f\}$ by

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Simulated annealing (SA) Derived as an analogue of annealing in metallurgy, simulated annealing (SA) uses change in pixel label over iterations and estimates the difference in energy of each newly formed graph to the initial data. If the newly formed graph is more profitable, in terms of low energy cost, given by:

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Supervised image segmentation using MRF and MAP In terms of image segmentation, the function that MRFs seek to maximize is the probability of identifying a labelling scheme given a particular set of features are detected in the image. This is a restatement of the maximum a posteriori estimation method.

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$$P(\lambda) = \sum_{\lambda \in \Lambda} P(\lambda \mid f) \mid \Omega \mid \quad \{\displaystyle P(\lambda) = \frac{\sum_{\lambda \in \Lambda} P(\lambda \mid f)}{|\Omega|}\}$$